Renaissance Philanthropy | SPRIN-D

BiTS SPRIND Fellowship 2026
Concept Note

Francesca Castaldo • Co-founder and Executive Director • Barcelona Computational Foundation (BCOM) • francesca.castaldo@bcom.one

ANPSY | ALGORITHMIC NEUROPSYCHIATRY

Personalised mechanistic models for predictive mental-health care

How does subjective experience arise from brain function, and how can that knowledge improve mental-health care? Without an account of the processes shaping experience, treatment remains largely trial and error. ANPSY will build a clinical system around personalised models connecting brain dynamics, behaviour and lived experience to estimate how clinically appropriate options are likely to affect an individual before treatment begins.

Programme objective

> **Build a clinical decision-support system that uses patient-specific computational models to forecast how alternative treatments affect brain dynamics, behaviour and lived experience. Validate its predictions and demonstrate its safe, practical use in care, beginning with major depressive disorder.**

ANPSY builds an end-to-end care system around personalised mechanistic digital twins connecting brain-circuit dynamics, cognitive functions and lived experience. Each model represents an embodied person in their social and environmental context, including relevant influences on symptoms and treatment response. It provides executable, falsifiable forecasts of benefit, burden, adverse effects and experiential impact under alternative interventions.

The first clinical evaluation focuses on neuromodulation, with psychotherapy, medication and possible combinations in scope; the population, interventions and comparator will be agreed during programme design. Within this clinical setting, models seek the processes generating individual symptoms and treatment response, including mechanistic distinctions obscured by diagnostic categories. Testable models of brain function, symptoms and reported subjective experience provide the scientific basis for this clinical capability and reusable methods for other indications.

THE PROGRAMME AT A GLANCE **€45M** 5-year target

PREPARATORY PROGRAMME DESIGN Months 0–3 **€250K**

Evidence review and expert meeting; costed implementation plan.

I DATA INFRASTRUCTURE €4.75M	II MODELS RANKED €10M	III VALIDATION & PILOT €30M
YEAR 1	YEAR 2	YEARS 3–5 TARGET
Can we build a shared evidence base?	**Which models predict best?**	**Can models support care?**
Existing data harmonisation; secure ingestion of new clinical measurements and outcomes.	Funded model development, competitions on common tests and independent qualification.	**IIIa Shadow mode** Test forecasts hidden from care teams. **IIIb Model-guided pilot** Test predictions, feasibility and safety.

Current practice and its limitations

> **Current mental-health diagnosis and care are based on symptoms and population evidence. Clinicians personalise thoughtfully, but care still lacks an individual account of how symptoms and subjective experience arise and persist. Effective treatment is therefore discovered largely through trial and error.**

Clinical teams assess symptoms, functioning, risk, physical health, history and social context. They select psychological, pharmacological, social or neurostimulation treatments based on clinical guidelines, previous treatment response, potential side effects, patient preferences and access to care. Medicines and devices are developed for diagnostic indications and validated through group-level trials. The strongest baseline ANPSY must beat is **high-quality, guideline-concordant, measurement-based care with shared decision-making and timely adjustment**.

Limitations. Diagnostic categories organise care but do not reveal the underlying mechanism maintaining illness in a specific person. Group-level estimates do not, by themselves, provide patient-specific forecasts of response, timing, mechanism or lived experience. Visits and rating scales sample a dynamic illness intermittently, while context, adherence, side effects and symptoms evolve between appointments. Personalisation therefore becomes more precise as treatment unfolds—sometimes only after months of ineffective exposure, adverse effects or relapse.

The best emerging approaches. Measurement-based care, pharmacogenomics, digital phenotyping, precision neuroimaging, circuit-targeted interventions and AI decision-support are reducing parts of this uncertainty. They show gains in speed, treatment continuation and targeting, but remain heterogeneous, difficult to transfer and unevenly validated[1].

Purely data-driven AI often needs large, hard-to-assemble datasets and can be difficult to interpret. Without explicit mechanisms, its use for simulating interventions is also limited; biologically grounded structure can constrain the search space.

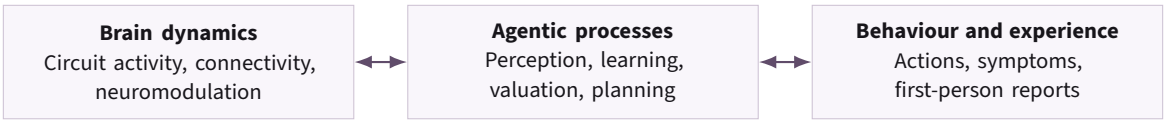
> **The unresolved gap.** Before treatment begins, care lacks a patient-specific mechanistic account of how brain function, body and context give rise to what a person feels, why distress persists, and how a particular intervention could change those processes. ANPSY links this account to testable predictions of neural, behavioural and experiential change.

[1] Husain et al., JAMA Netw Open 2025; Cipriani et al., JAMA 2026; Blake et al., JAMA Netw Open 2026; Esmaeilian et al., Psychiatry Clin Neurosci 2026; Leaning et al., Neurosci Biobehav Rev 2024; Siddiqi et al., Neuropsychopharmacology 2024.

Scientific approach and basis for success

ANPSY will build a reusable clinical system around patient-specific, mechanistic models that connect brain-circuit dynamics to behaviour and lived experience and can simulate alternative treatments.

The algorithmic leap. ANPSY represents the person as an embodied agent whose brain, body, beliefs, goals, actions, environment and subjective experience evolve together. The agentic model provides a conceptual scaffold for modelling brain circuits and dynamics as the biological implementation of perception, learning, valuation, goal formation, planning and action. These functions provide the bridge to behaviour and lived experience.



Experience enters the model. Through algorithmic modelling, ANPSY will link neural and clinical measurements with structured reports of lived experience. Interviews, speech and text provide first-person evidence, with coding and uncertainty retained. Paired observations constrain the model and test predictions of specified, reportable aspects of experience and treatment response; reports do not exhaust experience itself.

Why now. Advances in AI, shared multimodal datasets and scalable computing, alongside an established interdisciplinary talent pool, make coordinated testing practical. MRI, fMRI and EEG characterise brain structure and dynamics; tES and TMS provide measurable dose and circuit-engagement evidence[2].

Barriers to integration. Models of experience, interoception, decision-making and circuits remain fragmented. The unmet need is a clinically validated decision-support framework linking first-person experience with third-person measurements to forecast individual treatment response. ANPSY addresses this through paired observations, common interfaces and prospective intervention tests[3].

Open on architecture; strict on capability. ANPSY invites world-leading groups with complementary methods to develop alternative twins. Teams share data and clinical infrastructure under active management; models compete through independent tests. Teams may combine biophysical, Bayesian, dynamical, machine-learning or hybrid methods within a patient-specific architecture linking brain dynamics to behaviour and experience. Each live system must include explicit circuit or whole-brain dynamics, a defined treatment operator, uncertainty and prospective testability; strong data-driven predictors run in parallel as frozen challengers. Neural prediction, treatment-specific prediction and clinical benefit require separate tests.

Scientific foundation and delivery experience. Francesca Castaldo, BCOM's co-founder and executive director, convenes ANPSY and coordinates the €8M Horizon Europe Health project SYNERGIA. Karl Friston, Xiaosi Gu and Michael Fox advise the programme; Greg Witkop is its programme mentor. Giulio Ruffini (ERC GALVANI), Viktor Jirsa (The Virtual Brain) and Gustavo Deco (ERC NEMESIS) bring modelling and translational expertise[4]; Gitte Knudsen and Michael Nitsche bring clinical and neuromodulation expertise. This expertise provides a base for convening the programme; institutional participation and delivery roles will be agreed during programme design. BCOM's partnerships can support community engagement, outreach and translation into practice.

These assets provide simulation tools, personalised stimulation methods and experience linking brain dynamics to function. SYNERGIA's study design and clinical-delivery experience inform ANPSY's measurement and modelling framework.

The change in care

ANPSY will bring individualised computational models into treatment decisions, linking brain function with behaviour and first-person experience to improve patient outcomes and extend the reach of mental-health care.

Overarching clinical ambition. In the first depression application, the ambition is 15 additional remissions per 100 offered model-assisted care at 12 weeks.

The programme links shared data, competing models and prospective clinical testing. Its five-year target is qualified predictions and a tested decision-support workflow.

Benefits in practice. Patients could recover sooner and avoid ineffective treatments; families could face less caregiving burden and disruption. Clinicians could gain individual treatment guidance, while health systems gain evidence on outcomes and resource use. Shared models, data and standards can extend these benefits to other disorders. At scale, sustained recovery could mean fewer years lost to illness, greater participation in work and society, and social and economic benefits that accumulate over decades.

[2] HCP Young Adult, data release 2025; EBRAINS, Computing Infrastructure (accessed 2026); Siddiqi et al., Neuropsychopharmacology 2024; Taylor et al., JAMA Psychiatry 2026; Jog et al., JAMA Netw Open 2025.

[3] Hesp et al., Neural Comput 2021; Harrison et al., Front Behav Neurosci 2025; Sadeghi et al., npj Ment Health Res 2024; Gu et al., Wellcome landscaping report 2023; Zierer et al., J Affect Disord 2024.

[4] The Virtual Brain, EBRAINS; ERC GALVANI, CORDIS 855109; ERC NEMESIS, CORDIS 101071900.

THE SCALE OF NEED: EU27

≈14.0M
people living with major depressive disorder

€3.25T
modelled economic output loss over 25 years

LONG-TERM CLINICAL VISION

26.3M
additional remission outcomes over 25 years

€171B
potential gross output recovered over 25 years

Illustrative 25-year scenario following demonstrated clinical benefit and adoption: fixed population and €130B annual loss; 50% reached each year, one treatment opportunity per reached person-year, 15 additional remissions per 100 at 12 weeks, and 70% economic restoration. Outcomes may include repeat patients. Amounts are undiscounted gross output before costs. Population and burden are OECD-derived approximations, with Eurostat population data.[5]

Enduring infrastructure and coordinated funding. ANPSY will leave a harmonised longitudinal data resource and reusable pipeline linking brain, body, behaviour, context and lived experience to patient-specific mechanistic models and treatment simulation. This shared work exceeds an ordinary project's scale and remains too scientifically ambitious for conventional commercial investment at this stage. Philanthropy and mission-driven public funding can support its coordinated development, independent evaluation and clinical transition.

Responsible modelling and clinical action

> **A system built to understand a person can cause harm if observation becomes surveillance, a forecast becomes an identity, unequal data become unequal care, or an overconfident recommendation directs the wrong intervention.**

Scientific risk. Models may reproduce existing observations yet fail to predict treatment response. ANPSY tests the proposed experience–brain links through prospective forecasts and comparisons with equally informed models that retain the measurements but remove the declared link.

Protect the person. Longitudinal does not mean always-on. Collection is purpose-bound, minimal, transparent and controllable; people can pause or withdraw without losing ordinary care. Linked contextual data, including data about others, are treated as potentially identifiable. Monitoring burden, behavioural reactivity and unwanted self-focus are measured as possible harms.

Protect the decision. Models expose uncertainty and can abstain. In clinical use, patient and clinician define admissible options and select treatment; the clinician authorises delivery. The patient may decline any option. Disagreement with first-person evidence triggers review. Interface design addresses nocebo, self-fulfilling and automation effects.

Protect fairness and society. Performance, drift, access and burden are audited across populations and sites. Outputs are barred from employment, insurance, marketing, social scoring or other secondary use. Cybersecurity, third-party privacy and digital exclusion are programme-level safety targets.

Staged funding requirements

> **Immediate request: €250K for three months of programme design, included in the €5M first year. Proposed programme envelope: €45M for programme design and Phases I–III, targeting five years through pilot analysis.**

Different funders may support successive stages. The phase allocations are shown in the overview. Pilot scope, duration and continuing team costs require assessment.

Programme timeline

> **Five-year programme target through pilot analysis, subject to the costed clinical protocol.**

Preparatory programme design runs in Months 0–3; Phase I establishes the shared infrastructure (Months 3–12). Phase II funds model development and independent qualification (Year 2). Phase III validates hidden forecasts and admits qualified systems to a separately gated pilot. The costed protocol must establish whether clinical follow-up and analysis fit the five-year target; evidence and readiness determine any overlap.

Milestones and success criteria

Milestones mark accepted releases. The database grows through validation and the pilot; ingestion and curation continue thereafter. Dates depend on independent review at each gate.

Preparatory programme design | month 3. Agreed dossier: clinical use case, measurements, data-access plan, benchmarks and protocol synopsis.

I Data infrastructure | month 12. Operational infrastructure with harmonised initial data, tested ingestion, reproducible scoring and a costed clinical protocol; evaluation rules fixed.

II Models ranked | month 24. Independent neural, treatment-specific and experience–brain tests rank models and qualify their claims; treatment rankings require comparative evidence and sites are ready. Candidate target: 10% lower forecast loss than a strong clinical-only reference; the margin and confidence rule are fixed before testing.

IIIa Shadow validation | evidence gate. Frozen forecasts are evaluated on new patients; independent evidence and clinical readiness determine pilot entry.

IIIb Pilot and analysis | month 60 target. Complete the agreed clinical follow-up and independent prediction, safety and feasibility analyses; release the governed pilot resource and recommendations for future clinical development.

[5] OECD, The Economic Case for Preventing Mental Ill Health, 2026; Eurostat, EU population continued to grow in 2025, 10 July 2026.

ANPSY
Algorithmic neuropsychiatry

Appendix

Programme and evidence

Supporting material for internal planning and selected funder discussions.

CONTENTS

A1	Scientific explanation: from brain dynamics to experience	5
A2	European landscape and ANPSY's contribution	6
A3	Funding, governance and programme overview	7
A4	Preparatory programme design	9
A5	Phase I: database infrastructure and measurement foundation	10
A6	Phase II: competing models and independent qualification	11
A7	Phase IIIa: prospective hidden forecasts	12
A8	Phase IIIb: decision-support pilot	13
A9	KOKORO: measuring lived experience	14
A10	KAIROS: hormone-informed mental-health forecasts	15
A11	Data access, privacy and the regulatory route	16
A12	Bottlenecks: owners, decisions and alternatives	17
A13	Economic impact: value and assumptions	18
A14	Why now: a stronger scientific foundation	19
	Selected references	20

Francesca Castaldo
Barcelona Computational Foundation (BCOM)
francesca.castaldo@bcom.one

26 October 2026

A1. Scientific explanation: from brain dynamics to experience

AIM

Explain how brain mechanisms shape a person's cognition, behaviour and lived experience, and forecast how treatment changes this linked system. ANPSY develops executable patient models whose neural, experiential and clinical predictions can be tested separately and together.

The person as an embodied agent. Our algorithmic-agent account describes how a person builds a model of themselves and their world, evaluates possible states through affective valence, and plans actions. Depression can involve different disturbances in these interacting functions: persistently negative expectations, reduced reward valuation, difficulty imagining effective actions, or adverse circumstances.[18] ANPSY connects these computational functions to their neural implementation. A patient model must specify how changes in circuit dynamics alter learning, valuation or planning, and how these changes appear in behaviour and reported experience. Each model must specify which bodily and contextual variables it uses and how they affect the modelled processes.

From neural populations to measurements. A whole-brain model is one concrete route. Neural masses describe interacting populations of excitatory and inhibitory neurons; anatomical connections couple regions through modelled weights and delays. Structural and diffusion MRI constrain anatomy and pathways, while EEG, MEG or functional MRI constrain dynamics.[19,20] Observation models translate simulated sources into electrical or magnetic signals, or into fMRI's indirect blood-oxygen signal. This distinguishes hidden neural processes from the measurements used to estimate them.

Learn across people; infer within the person. Development data supply shared representations, dynamical structure and physiological priors. For a new patient, a fixed inference procedure estimates individual parameters and current state from the permitted observations, retaining uncertainty. Anatomy, physiological parameters and initial state are distinct sources of personalisation. AI can support feature extraction, parameter inference and fast approximations of costly simulations; mechanistic equations specify how states evolve. Later observations test both the frozen model and its personalisation procedure. Recovery in simulations, accuracy on human observations and computational efficiency have separate checks.

Treatments change specified processes. Treatment operators represent an intervention's target, dose and timescale. Drugs may alter receptor-dependent circuit responsiveness; electrical stimulation enters through the field estimated from a head model. A receptor-informed whole-brain model has reproduced aspects of LSD-related functional imaging through changes in neuronal gain.[21] Longer-term forecasts also need adaptation and learning where relevant. Psychotherapy operators would specify changes in learning, beliefs or action strategies and test their consequences. Each operator connects a proposed mechanism to measurable responses under that intervention.

Measure experience in the person's own words. Repeated speech and text reports preserve what matters to the person: felt mood, expectations, agency, bodily experience and everyday concerns. Natural-language methods can represent meaning as quantitative features alongside voice measures, while retaining the original account.[22] Semantic measurement studies support this route beyond fixed questionnaire items.[23] Participant judgments, blinded ratings and established scales test whether representations preserve meaning, remain stable when experience is unchanged and detect meaningful change. Language, context, recording conditions and missing reports are part of measurement validation. KOKORO develops this layer (Section A9).

Pair experience with brain, behaviour and body. Time-aligned reports, cognitive tasks and neural recordings let the model predict linked trajectories within the same person. A proposed mechanism of reduced reward valuation should account for relevant neural/task responses and changes in motivation described in daily life. For example, a circuit account of depression relates persistent negative valence to dysfunction in distributed valuation circuits, including the subgenual anterior cingulate cortex (sgACC).[18] Deep brain stimulation (DBS) studies have stimulated nearby white-matter pathways; repetitive transcranial magnetic stimulation (rTMS) targets connected prefrontal cortex.[24,25] Such a model could link stimulation-induced changes in circuit coupling to sensitivity to negative outcomes, predicting task performance and reported mood. Paired neural, behavioural and experiential measurements would test this proposed mapping. Observation models make these mappings explicit, including uncertainty. Repeated measurements distinguish stable differences between people from changing states within a person. Human studies of cortisol rhythms and ovarian-hormone variation motivate additional bodily measurements;[26,27] KAIROS tests hormone sensitivity as a bounded extension (Section A10). Pairing and temporal order are essential to testing the proposed links.

Testing the contribution of mechanistic coupling. Mechanistic coupling defines ANPSY's scientific objective: explain individual processes and simulate specified interventions. Neural modelling and computational accounts of cognition provide the starting evidence; their integration is tested against equally informed alternatives. Compare the full model with fairly retrained versions that omit additional experience measurements or remove the explicit link while retaining the same inputs. Compare individual physiological parameters with population parameters while keeping the person's anatomy and opportunity to infer state. Strong AI-only and mechanistic-only references test any claimed hybrid advantage.

Independent tests use held-out people, later observations and external sites where available, with matched information cutoffs and fair tuning. Parameter identifiability and perturbation evidence support biological interpretation. Neural prediction, experience prediction and treatment selection each have their own evidence requirements and qualification decisions (Sections A6–A8). This makes the proposed mechanism an experimentally testable part of the clinical system.

A2. European landscape and ANPSY's contribution

ANPSY proposes a shared scientific and clinical programme linking mechanistic brain–cognition–experience models, paired measurements and competing systems. Depression is the first application; the longer-term ambition is a reusable foundation for neuropsychiatry across European clinics. Existing programmes provide methods, infrastructure and clinical expertise on which to build.

Relevant programmes

PETRUSHKA. Clinical and demographic information, treatment evidence and patient preferences inform antidepressant choice. Its multinational randomized trial reduced eight-week discontinuation (17.0% versus 27.4%). Unmasked treatment and substantial missing symptom data limit interpretation of later outcomes. It is a methodological reference for clinical decision support; ANPSY's intervention-specific comparators will follow the agreed clinical question.[1]

OPADE. This €10M programme combines multi-omics, wearable EEG and cognitive/emotional measurements to predict antidepressant response. Its 350-patient protocol is observational. ANPSY adds an explicit model of the coupling between brain dynamics, cognition and lived experience, with separate tests of treatment selection.[2]

Psych-STRATA, PSY-PGx and REALMENT. European precision-prescribing programmes combine biological data, algorithms and clinical implementation. Psych-STRATA runs treatment-strategy trials, including depression; PSY-PGx tests genotype-guided dosing in a randomized trial; REALMENT develops prediction and the 4MENT clinical platform. These offer comparator methods and potential clinical partnerships.[3]

Virtual Brain Twin. The €10M EBRAINS-coordinated project develops multiscale models for schizophrenia treatment, including medication and stimulation, with prospective studies planned. ANPSY starts in depression and proposes paired neural, cognitive and experiential measurements, competing models and a staged programme from independent qualification to a clinical pilot. The shared mechanistic foundation creates opportunities for collaboration.[4]

SYNERGIA and mechanistic foundations. SYNERGIA plans a study of psilocybin plus active/sham tDCS in depression with whole-brain modelling. Neurotwin and GALVANI develop personalised stimulation models; PRISM2 links neural networks and social function. ANPSY brings such methods into a continuing programme of comparative clinical qualification.[5]

ANPSY's additional contribution

An explicit scientific hypothesis. The executable connection between brain dynamics, cognitive processes and lived experience is central. Paired observations constrain the model; component tests establish whether this coupling improves treatment predictions beyond strong alternatives using the same inputs.

A programme for competing systems. Shared acquisition protocols, data and clinical infrastructure support competing teams under sustained programme management. Concealed forecasts, frozen models, fair comparators and independent gates determine progression. The output includes a clinical system and a continuing route for qualifying new approaches.

Ambition with defined clinical tests. Each treatment comparison is separately qualified, beginning with a defined depression decision. Forecasting and treatment selection face distinct tests; the pilot assesses prediction performance, safety and practical use.

Clinical pilot across European sites. Common protocols, external validation, clinician training and local care integration support the agreed pilot. Potential collaborations include EBRAINS for modelling and federated data services, ECRIN for multinational trial operations, and TEF-Health for testing in clinical settings. Programme design will develop the practical arrangements with prospective partners.[6]

Selected primary sources

1. PETRUSHKA randomized trial, JAMA 2026; tool development.
2. OPADE, CORDIS 101095436; clinical protocol.
3. Psych-STRATA; INTENSIFY methods; PSY-PGx trial protocol; REALMENT.
4. Virtual Brain Twin, CORDIS 101137289; clinical milestones; EBRAINS: clinical plans.
5. SYNERGIA; Neurotwin; GALVANI; PRISM2.
6. EBRAINS MIP; EBRAINS simulation; ECRIN; TEF-Health.

A3. Funding, governance and programme overview

> **Immediate invitation: €250K for programme design.** This produces a costed, independently reviewed implementation dossier within three months. Later funding follows evidence gates and agreed responsibilities. The proposed envelope is **€45M for preparatory programme design and Phases I–III**.

Phase	Timing	Envelope	Deliverable
Design	Months 0–3	€0.25M	Programme design and implementation dossier.
I	Months 3–12	€4.75M	Authorised data, measurements, reference environment and protocol.
II	Months 12–24	€10M	Competing systems, independent qualification and clinical readiness.
IIIa / IIIb	Years 3–5 target	€30M	Hidden forecasts, separately gated pilots, final analysis and continuing model support.

Budget boundaries. Year 1 totals €5M; the programme envelope is €45M. IIIa and IIIb share €30M. Cost recruitment, follow-up, model-team continuation, integration, shared services and evaluation before commitment. Study-specific costs are assigned once to Phase I/II/III work packages or separately financed. Include participant involvement, rights, indirect costs and transitions. KOKORO and KAIROS retain distinct funding scopes; distinguish new work from existing contributions.

A continuing programme across funding stages

Philanthropy can support launch, integration and the core team across grant cycles. Programme management maintains common questions, interfaces and evaluation rules; delivery responsibilities and payment arrangements will be agreed during programme design. Clinical and data obligations continue when a model does not advance.

Accountability and independence

BCOM proposes to convene and coordinate ANPSY, with co-coordination arrangements considered during programme design. Programme design assigns the programme operator, clinical lead and sponsor route, resource provider, measurement lead and independent evaluator. Patient contributors help set outcomes, burden and acceptable uses. Appointments and institutional participation remain subject to agreement.

The evaluator controls final tests and scoring. An independent panel selects proposals on scientific merit and recommends progression; the agreed decision authority approves awards and progression under published rules. Teams receive equivalent authorised development information; final-test access excludes competition on that test. Decisions are recorded, with declared interests and recusal. Contributors cannot assess their own submissions or competing awards.

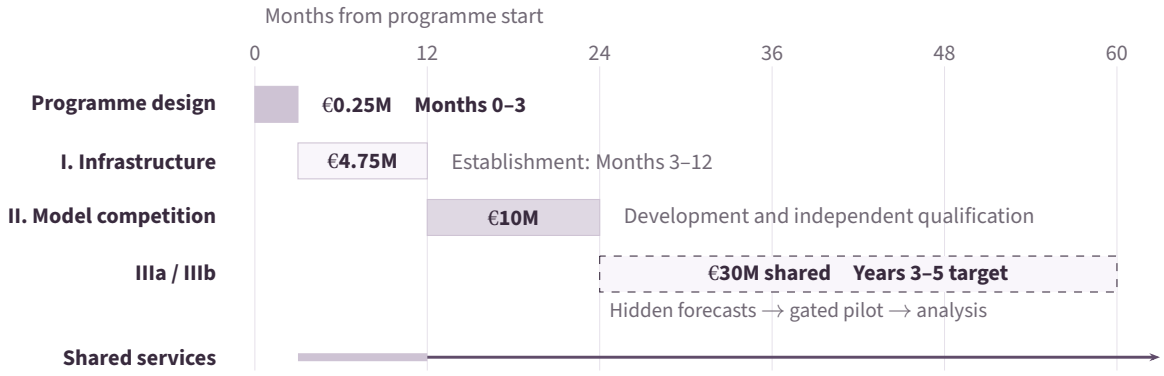
Roles, collaboration and lasting access

Francesca Castaldo is BCOM's co-founder and executive director. She leads ANPSY and coordinates SYNERGIA from BCOM. She is also currently a scientific programme manager and senior researcher at Neuroelectrics, a manufacturer of transcranial electrical stimulation devices, including tDCS. Programme coordination, model development and independent evaluation have distinct responsibilities under the same governance rules for all contributors.

Awards will define background IP, permitted execution and reuse, publication, reproducibility and handover. A long-term host and costed service agreement must sustain accepted resources after the programme, subject to custodian rights. Claims, funding and timelines are revised when a gate is failed or inconclusive; essential clinical responsibilities continue through those decisions.

Programme timeline and Phase II target

Three-phase programme: €45M, targeting five years through pilot analysis. Timing remains subject to the costed clinical protocol and evidence readiness. Proposal selection and agreements precede funded model development.



Shared infrastructure is established in Phase I, maintained throughout the programme and designed for use beyond it. Continuing services are costed across the relevant phases and participating projects.

Pilot duration within the programme target
Evidence and readiness gate → **at least 24 clinical months**, including final follow-up → analysis and independent review.
With 24 clinical pilot months and six months for analysis and review, completion by month 60 requires pilot activation by month 30. Phase IIIa must therefore provide the evidence needed for pilot entry within six months of its month-24 start, while longer follow-up may continue. Programme design will test this schedule against recruitment, outcome timing and review requirements, and revise the completion date if necessary.

What the Phase II performance target means

Candidate target: 10% lower treatment-outcome forecast loss than a strong clinical-only reference, measured on the same protected, unseen patients and outcome.

$$\text{Relative gain (\%)} = 100 \times \left(1 - \frac{\text{model loss}}{\text{clinical-reference loss}}\right).$$

Illustration: a reference loss of 100 and model loss of 90 gives a 10% relative gain. This illustrates the calculation, not an observed ANPSY result.
Phase I fixes or revises the margin and confidence rule using reliability and precision evidence, before protected testing. Report absolute and relative gains, uncertainty, calibration and errors in the original outcome units; compare an equally informed flexible model alongside the clinical reference. A point estimate alone does not establish qualification.
This target concerns outcome forecasting. Neural and joint brain–experience tests have task-specific scores and margins (A6). Treatment rankings require protected comparative evidence; forecasting performance alone cannot qualify them.

The conditional vision for clinical deployment is in Section A8.

A4. Preparatory programme design

> **Months 0–3**
> **€250K within Year 1**
> **Deliverable:** An implementation dossier ready for a Phase I award

Purpose

Turn the scientific programme into an agreed, costed specification for delivery. The work resolves the choices that need clinical, modelling, statistical and operational agreement: the first treatment decision, minimum measurements, usable evidence, evaluation rules and responsibilities. The award purchases a completed implementation dossier.

Scientific convening and invited expertise

Programme advisers Karl Friston, Xiaosi Gu and Michael Fox, with programme mentor Greg Witkop, are helping shape the scientific convening on computationally driven therapeutics for depression. Participation and delivery roles will be agreed with each contributor and institution. The group will combine thinkers who can frame the scientific questions with specialists who can turn them into executable studies: clinical psychiatry, mechanistic and cognitive modelling, pharmacology, neuromodulation, experience measurement, statistics, data governance and trial delivery. Patient contributors bring priorities and practical constraints.

The meeting will produce agreed methods and specifications that ANPSY can implement and that can serve as a reference for other neuropsychiatry programmes of comparable scope.

Work and deliverable

The three-month design phase combines preparation, scientific convening, synthesis and independent review. The programme design will set the format, invitation list and working-group responsibilities.

Working groups deliver five linked parts of the **Computational Neuropsychiatry Programme Dossier**:

- **Clinical decision.** Agree the first neuromodulation decision for depression, population, admissible options, preferences, outcome and horizon. Specify the TMS/tES target, parameters, therapy or combination question against clinical relevance, model operators and usable evidence. DBS requires a separately scoped invasive specialist study.
- **Measurement and model contract.** Define permitted inputs, paired experience and neural measurements, acquisition timing, intervention descriptions and predictive outputs. Specify a feasible core battery and justify any intensive subset. Fix the scientific comparisons that test the contribution of physiology and experience.
- **Evidence and access route.** Identify data custodians and rights holders, count eligible records, and specify prospective acquisition and alternatives for unsuitable existing data. Assign responsibility for permissions, a clinical sponsor route and provider assessment. Record the access and participation arrangements needed for each contribution.
- **Evaluation specification.** Define independent execution, protected tests, fair reference models, uncertainty and missing-input rules. Set the process for establishing useful-gain margins from Phase I reliability, baseline and precision work before confirmatory evaluation.
- **Delivery and financing.** Produce a clinical synopsis, work packages, accountable functions, milestones and a costed Phase I plan. Include proposal review, award preparation, continuing model-team work, pilot duration and analysis. Identify shared services, continuing obligations and boundaries with existing grants and separately funded applications.

Evidence, timing and cost

The dossier includes an evidence register and dataset feasibility matrix, contributor decisions, a paired measurement specification, benchmark plan, clinical synopsis and costs. Each unresolved dependency has an owner, deadline and consequence for scope.

The €250K planning envelope covers preparation, expert and patient involvement, convening, synthesis and independent review. The costed plan will confirm the format, supplier prices, eligible honoraria and staffing before award; data engineering and full protocol development belong to Phase I.

Progression decision

Independent acceptance asks whether the dossier supports an executable Phase I award with a credible clinical and data route. Payment can follow accepted deliverables. If a major route is unresolved, fund the bounded work needed to resolve it, revise scope or defer the dependent activity.

A5. Phase I: database infrastructure and measurement foundation

Establishment: Months 3–12
€4.75M
Deliverable: A governed, reusable data and measurement infrastructure and Clinical Protocol v1

Shared infrastructure throughout the programme and beyond

Phase I establishes shared data and measurement infrastructure for ANPSY's depression programme. It supports the full programme and is designed for continued operation and reuse beyond it. Related projects, including hormone-informed work such as KAIROS (Section A10), may share and help develop its services under agreed responsibilities, access permissions and cost allocations. The deliverable is an operating research resource: authorised teams can execute models on documented records, and clinical sites can contribute new measurements through a tested process.

Governance and stewardship. BCOM is engaging experts in regulatory affairs and data protection to guide the governance design. Appoint an accountable Reference Resource Provider, data lead and independent evaluator. Clinical, scientific, stewardship and patient/citizen representatives oversee access, reuse and collection burden. Custodians retain control under agreed rights; the evaluator controls protected tests. A11 specifies responsibilities and the access route.

Connect existing data pools. Assess collaboration and integration before commissioning new services. EBRAINS HealthDataCloud offers a federated setting for sensitive research data; its Medical Informatics Platform supports analysis across clinical data held at source. REALMENT supplies relevant experience in harmonised psychiatric data and distributed analysis. Assess their fit for ANPSY's paired measurements and executable models, alongside the programme resources in A2. Agreements will specify usable records, services and continuing costs.

Candidate resources for assessment. BRIGhTMIND, PDC, PReDICT, EMBARC and CAN-BIND-1. Phase I checks task fit, paired measurements and permissions for development and independent evaluation; no access is presumed.

Data ready for analysis and ingestion

Linked, task-specific releases. Connect permitted pretreatment measurements, treatment allocation/exposure and later outcomes. Count usable records by task and treatment arm, including timing, missingness and paired neural–experience coverage. Confirm development, evaluator and reuse rights.

Documented harmonisation. Preserve source variables, units, provenance, cohort differences and non-equivalent horizons. Distinguish observed, inferred and imputed values. Fit transformations within development data. AI may propose mappings, extract time-stamped information and flag anomalies; experts check errors and retained meaning against source records. Missing first-person reports remain missing.

Prospective ingestion. Deliver identifiers, timestamps, device/site metadata, quality checks, missing-data codes, access roles, audit trails and versioned transformations. Run a representative site batch from acquisition to scored record. Test privacy transformations for effects on scientific usefulness. Support central or federated storage according to agreed permissions.

One interface, different architectures

Each model receives an eligible patient record D_i, treatment description a and horizon t, and returns forecast distributions for supported clinical, experiential and neural/behavioural targets:

$$(D_i, a, t) \longmapsto \{F^C_{i,a,t}, F^E_{i,a,t}, F^N_{i,a,t}\}.$$

Specify measurement definitions, information cutoffs and uncertainty. Publish versioned reference pipelines and scoring. The clinical-only reference retains routine patient reports and preferences; an equally informed flexible reference uses the same permitted measurements as the candidate. A universal simulator or new foundation model is not required.

Learn across people; infer within the person. Population training learns reusable structure and priors. A fixed inference procedure estimates a new person's physiology and hidden state from permitted inputs. Document anatomy, physiological parameters and initial state separately. Freeze the shared model and inference procedure before confirmation; neither future outcomes nor other test participants may update them.

Paired measurement and clinical design. Align first-person reports with neural, behavioural and clinical observations. Retain original reports and coding uncertainty; test meaning, reliability, responsiveness, completion and burden. Clinical Protocol v1 specifies hidden forecasting, pilot entry, overlap, follow-up and final analysis, recruitment, missing-data analyses, approvals and costs. Include the randomized evidence needed for Phase II selection claims and acquisition for gaps in legacy data.

Delivery, cost and continuity

M6: permissions, counts, schema and ingestion. **M9:** harmonisation, reference/site tests and precision/cost review. **M12:** reproduced scoring, protected tests and costed clinical design. Prepare invitations, independent proposal selection and agreements before funded development; awards remain subject to readiness.

The €4.75M envelope covers access and harmonisation, secure platform and ingestion, AI-assisted preparation, measurement and clinical design, benchmarks and evaluation, coordination and contingency. Price the first decision, selection preparation and recurrent services before allocating the envelope and committing funds. Phase I funds establishment and initial operation; continuing services are costed across later programme phases and participating projects. Agree a funded host/handover route for continued use, with FAIR metadata and explicit access terms.[28]

Gate I. Accept demonstrated permissions, paired coverage, reproducible execution, tested ingestion and clinical feasibility. Approve an informative existing-data route or financed acquisition with sponsor, approvals schedule and achievable completion. Narrow the claim or revise timing where necessary. Supported forecasting may continue; treatment selection awaits the protected comparative test.

A6. Phase II: competing models and independent qualification

Months 12–24, subject to evidence readiness
€10M
Deliverable: Qualified claims, reproducible model dossiers and clinical readiness

Select proposals, fund development, then compete

Groups submit proposed approaches and delivery plans, using existing assets to demonstrate feasibility. An independent panel selects proposals on scientific fit, feasibility, capability, potential and complementarity. The programme decision authority approves awards under published rules. Entry and award numbers will follow the costed design. Funding precedes new development; substantial implementation requested before award needs a paid scope. Contributors cannot assess their own submissions or competing awards.

Selected teams build, improve and integrate their systems, then freeze them to compete on common protected tests and undergo independent qualification. Rank alone does not permit clinical progression. Biophysical, Bayesian, dynamical, machine-learning and hybrid architectures may meet A1; strong AI-only references run alongside them. The €10M covers model awards, shared evaluation, integration and clinical readiness. Entry requires an ethics/privacy and service-access plan. Independent reviews assess continuation; shared clinical/data services have separate decisions.

Common tasks, separate evidence

Neural prediction. Infer an unseen person's model from allowed recordings; predict features in a later withheld window. Phase I fixes modality, reliable targets, cutoffs, the interval separating input and evaluation windows and gain margin. Compare population, persistence, temporal and strong AI-only/mechanistic references. Exclude target access and future-window filtering or normalisation.

Joint scientific tests. Test paired brain–behaviour–experience predictions, added experience inputs and cross-domain coupling. Compare personal versus population physiology with the same individual anatomy, observations and state inference; retrain comparators fairly. Distinguish simulator fidelity, parameter recovery and prediction of human observations.

A: outcome forecasting. Predict fixed-horizon outcomes from pretreatment inputs and treatment. Assess Brier scores for binary outcomes and continuous ranked probability scores (CRPS) for continuous outcomes, calibration and errors in the original outcome units. Phase I fixes or revises the candidate 10% forecast-loss reduction using reliability/precision evidence and a prespecified confidence rule. This is not a remission gain; prognosis alone cannot select treatment.[29] The target is illustrated on page 8.

B: treatment selection. Freeze option-specific forecasts and a selection rule. In protected randomized comparisons, estimate extra remissions per 100 using allocation probabilities and a prespecified inverse-probability or augmented estimator. Compare a development-selected fixed option, a strong clinical/preference rule and an equally informed flexible model. Prespecify margin, confidence, missingness and multiplicity; examine rule disagreement and counts by arm. Imprecise gains remain inconclusive.[30]

Independent execution. Common interfaces, fair tuning, participant separation, external-site tests where feasible, limited feedback and checks for test-data exposure during pretraining. Freeze code, preprocessing, personalisation, uncertainty and missing-input fallback. Report task/component scores, external performance, coverage, acquisition burden, runtime/compute and reproducibility. Never reuse confirmation data for tuning or track selection.

First treatment decision and data route

Clinical scope. Begin with model-informed neuromodulation for major depressive disorder, including non-invasive TMS/tES, with therapy and combinations specified during programme design. Agree the population, admissible target/parameter or treatment options, comparator, measurements, outcomes and horizons. DBS requires a separately scoped invasive specialist study. No protocol or dataset is presumed suitable before review.

Evidence route. Audit existing intervention datasets for permissions, linked records, treatment protocols and permitted inputs. The clinical question determines whether existing randomized evidence supports selection claims or new acquisition is required. Different intervention questions may need separate studies.

Rules and comparators require their pretreatment inputs, admissibility, allocation and outcomes; every recommended option needs positive randomized allocation probability. Absent preferences require prospective collection for a preference-aware rule. Modelled-change claims also require paired neural/experience follow-up. Component cohorts cannot replace missing selection inputs or randomization.

Cost the evidence route before commitment

Cost suitable existing-data access and prospective acquisition separately. Include equipment and sessions, clinical/research staff, paired measurements, follow-up, secure execution, independent evaluation and continuing model support. Sample requirements follow the prespecified precision analysis.

M6: access/counts audit. **M9:** precision and site/service quotes. **M12:** approve a financed, executable route. Assign costs once to Phase I/II/III work packages, identifying displaced work, or secure separate finance. Acquisition must leave time for independent confirmation; revise the dependent gate when necessary.

Gate II: qualify claims, not a winner

Dossiers record supported claims, failed/inconclusive tests and scope limits. Multiple systems, or none, may pass. Several models can share a prospective cohort without creating treatment arms. Prepare approvals, sites and workflows in parallel; IIIa also requires operational readiness.

Without adequate comparative evidence, withhold selection qualification and review bounded forecasting/mechanistic work. Observational IIIa cannot close this gap; unsupported rankings cannot enter IIIb.

A7. Phase IIIa: prospective hidden forecasts

From M24, subject to evidence and readiness
Within the shared €30M Phase III envelope
Deliverable: A prospective validation report and paired resource

Purpose

Test qualified models on new patients during clinician-led care. Forecasts are registered before treatment and concealed from patients and care teams. This establishes performance in the clinical setting, including acquisition burden and generalisation across sites, before experimental outputs influence treatment decisions.

Clinical and model workflow

- Obtain consent and confirm eligibility, treatment history, goals, preferences and feasible options. Collect the declared baseline experience, clinical, behavioural and neural measurements using the fixed acquisition and processing procedures. Record relevant sex and hormonal variables with timing and missingness.
- Clinician and patient choose treatment independently of experimental outputs. Record alternatives, contraindications, availability and reasons for the choice. Ordinary clinical and safety information remains available to care teams.
- Each frozen system registers treatment-conditional outcome distributions and uncertainty before treatment starts. Other supported options may be forecast for research, but the outcomes under treatments not received cannot be scored directly. The independent evaluator controls the forecast record.
- Follow clinician-led care with monitoring and rescue procedures. Record treatment exposure, adherence, changes, concomitant care, adverse effects and repeated clinical, experience and neural/behavioural outcomes at the agreed times.
- Evaluate the prespecified forecasts, scientific comparisons, calibration, coverage and external-site performance. Report unavailable forecasts and missing patient outcomes separately, using the locked handling rules. One shared cohort can evaluate several systems.

Models cannot refit to confirmation outcomes. Treatment changes after registration follow a declared initial-care-strategy analysis or a separately specified update task with a new information cutoff. The original forecast remains in the record. Incidental findings, safety escalation and exceptional unmasking have protocol-defined procedures.

Evidence and delivery

Compare clinical, experiential and neural/behavioural forecasts against the clinical-only and equally informed flexible references established in Phase I. Apply the scientific comparisons in A1. Report performance by treatment and clinically relevant subgroup where the data support it, with uncertainty, acquisition time, completion and cost.

IIIa covers recruitment, paired acquisition, follow-up, forecasts, analysis and independent review. A mature, prespecified analysis may support a pilot for a defined clinical use while other IIIa studies or longer-term follow-up continue. Entry depends on evidence and readiness, not a calendar deadline.

IIIa and IIIb share €30M. Cost clinical delivery, acquisition, secure services, independent evaluation and continuing model-team operation, input checks and failure analysis. Follow-up and stewardship remain funded through pauses in model awards. Timing follows the dependencies shown in A3.

Gate IIIa and inference boundary

Deliver a paired resource, executable frozen systems and an independent validation report for the proposed clinical use. A mature, prespecified analysis must show the required predictive and scientific performance, feasible measurements, acceptable uncertainty and a justified pilot risk–benefit assessment. Entry requires the comparative evidence appropriate to the decision; other IIIa follow-up may continue.

Clinician-selected treatments can differ in patient severity, history and preferences. Successful forecasting under care received therefore does not qualify treatment ranking. Comparative claims retain the separate randomized-evidence requirement in A6. A forecast-qualified system may proceed only to a pilot scope supported by its evidence; a system failing required performance or operational criteria returns to development or stops.

A8. Phase IIIb: decision-support pilot

At least 24 clinical months, then analysis
Within the same €30M Phase III envelope
Deliverable: Prediction validation and tested workflow

Purpose and entry

Validate prespecified predictions and safe, practical use within a defined care decision. Independent review sets permitted populations and claims. Forecasts and treatment rankings require their respective evidence; uncertainty labels cannot qualify unsupported rankings.

The pilot requires a sponsor, the applicable approvals, consent, monitoring, stopping procedures and funded follow-up. The model version, interface and clinical workflow are recorded before entry. Clinicians and patients retain treatment choice.

Work and deliverable

Co-design the interface with clinicians and patients. Show admissible options, supported forecasts, uncertainty, missing inputs and reasons to abstain. The clinician discusses applicability with the patient and authorises treatment.

Maintain a decision record linking the model version, information available, outputs shown, patient preferences, recommendation acceptance or override, reasons and care delivered. Patients may decline an option without losing ordinary care. An override is a clinical decision to understand, rather than evidence of failure by itself.

Deliver user materials, training, the tested workflow, pilot data, independent prediction and implementation reports, a risk/control register. Reports can support publications and fundraising under agreed rights; further awards remain conditional. Document human–system interaction using DECIDE-AI guidance.[31]

Evidence required

Before recruitment, set acceptable limits and precision requirements for:

- forecast accuracy, calibration and uncertainty under care delivered;
- recruitment, completion, forecast availability and turnaround;
- understanding of uncertainty, applicability and reasons to abstain;
- patient and clinician burden, workflow fit, uptake and overrides;
- safety events, escalation, continuity of care and technical failures.

Prespecified prediction validation is separate from exploratory benefit estimates. Register forecasts before intervention and score outcomes under care received. The pilot cannot establish definitive remission benefit or rare-event safety. The evaluator controls release; later versions may use designated data but require fresh qualification.

Timing, cost and intervention scope

Allow at least 24 clinical months, including final follow-up, then an illustrative six months for data lock, independent analysis and review. Close recruitment accordingly. Activation requires use-specific evidence and readiness; preparation can overlap IIIa.

The costed protocol must establish whether the clinical work and final analysis fit the five-year target. Evidence and readiness determine whether any overlap is justified. If the required work does not fit, scope, duration and funding need an explicit decision. Cost clinical work, model support, analysis, subsequent development, integration, secure operation and independent evaluation within the shared €30M.

Each stimulation, therapy, combination or invasive specialist study needs its own population, evidence and delivery plan; qualification does not transfer automatically.

Gate IIIb and fallback

Independent review assesses pilot performance, burden, safety and the claims supported by the evidence. Revise and retest failed workflows; exclude unsupported claims. Funded follow-up and stewardship continue if a model exits.

Conditional vision for clinical deployment

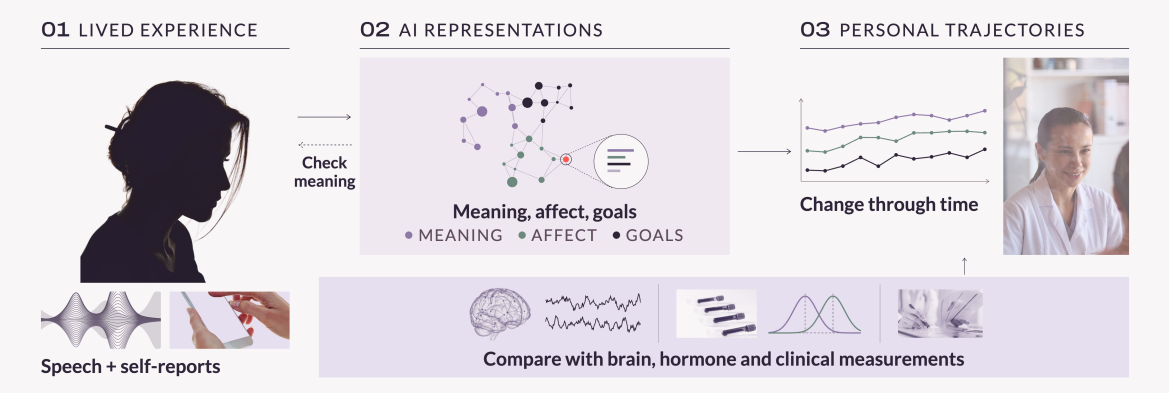
ANPSY's three-phase programme is a pathway to clinical deployment. Its shared data resource, qualified predictions and tested decision-support workflow provide the scientific and practical foundation for subsequent clinical development.

The longer-term clinical ambition is 15 additional remissions per 100 patients at 12 weeks, with fewer unsuccessful treatment attempts and less burden. This goal extends beyond the five-year programme; later clinical evaluation would establish the achievable benefit in the intended populations and care settings. If programme results support progression, further clinical evaluation and wider deployment would be separately scoped, approved and funded. Wider use would depend on demonstrated patient benefit, safety and value, clinical and regulatory readiness, delivery partners and continuing stewardship. The scale and form of subsequent development would follow the evidence generated by the programme.

A9. KOKORO: measuring lived experience

AIM

KOKORO will use language and speech models to quantify patients' accounts of experience beyond fixed questionnaire items, and test those measures against independent assessments.



Reports, representations and trajectories. AI-derived features remain traceable to the person's words and are compared with independent measurements. The workflow is proposed research; trajectories are schematic, not patient data or demonstrated clinical forecasts.

KOKORO makes a patient's account usable across visits while preserving the person's own words. A fixed guided interview records speech, produces a transcript and derives semantic and voice measures. The intended clinical research display places the account, its measurements and independent clinical assessments side by side. Within ANPSY, this could provide repeatable experience measurements for testing the connection between brain dynamics, cognition and clinical change.

Starting point and first funded output

BCOM has a working research-kiosk prototype with versioned session records. The proposed next step is a study application: reliable recording and export, longitudinal review, fixed scoring, quality flags and staff training. Each participant keeps their chosen native language across visits. The first demonstrator follows one person through two assessments and shows both accounts, derived measurements and independent ratings. Numerical scores remain traceable to the underlying report and scoring version.

A bounded measurement study

The proposed measurement study tests usability, transcription, meaning, stable-state repeatability and agreement across languages before clinical validation. Recruitment, clinical setting, sample size and follow-up will be specified in the separate study protocol, subject to capacity, costs and approvals.

Clinical validation will compare changes in prespecified experience measures with independent clinical assessments. The rater completes the clinical assessment before seeing KOKORO scores. Scoring and analysis are frozen before patient enrolment. Exploratory measures remain distinct from confirmatory tests. Pilot acceptance criteria, language effects, missing recordings and participant burden are specified before progression.

Funding and boundary with ANPSY

KOKORO is a separately costed research application covering software development, measurement validation and a reusable protocol. Funding, clinical deployment and validation remain pending. Incorporating its measurements into ANPSY requires an explicit allocation for integration and further validation. Shared development costs are identified once. SYNERGIA funding and recruitment are separate; any participation across studies requires its own arrangements.

Use, adoption and next decision

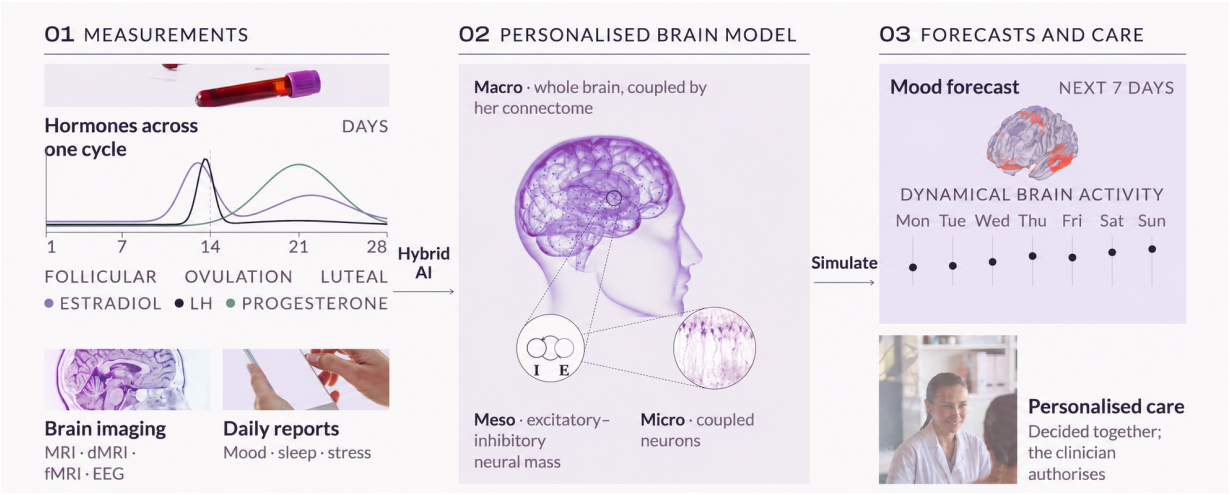
Patients, clinicians and study teams are intended users; research sponsors and trial sites are prospective purchasers of deployment, integration and support. The proposed release uses documented open-source terms, subject to third-party checks. Data permissions remain separate from software licences.

The next decision is to agree the study protocol, confirm delivery capacity and costs, complete the study build and agree pilot acceptance. Successful measurement validation would support reuse in other studies and an ANPSY integration test. Treatment selection, an autonomous clinical service and a commercial launch each require additional evidence, scope and funding.

A10. KAIROS: hormone-informed mental-health forecasts

AIM

Build a longitudinal dataset for understanding individual hormone sensitivity, and use it to develop personalised forecasts of mental health.



Hormone-informed whole-brain modelling. Sensor trajectories, imaging and daily reports would update a personalised model and its forecasts. Hormone coupling and clinical forecasting are proposed research. Curves and the seven-day horizon are illustrative.

KAIROS is a proposed application of ANPSY's modelling and evaluation approach to changes in women's mental health. Its first question is whether measured hormone trajectories improve forecasts of subsequent mood, sleep or symptom worsening beyond a person's previous reports, cycle timing and relevant context. A useful result would connect hormone sensing to a defined mental-health application and identify which measurements that application needs.

Define one forecast before expanding the study

The first funded package would agree one population, outcome and prediction horizon with prospective clinical and data partners. Naturally cycling adults with recurrent cycle-associated symptoms and a clinically characterised premenstrual dysphoric disorder cohort require different protocols. A next-day mood forecast is a candidate demonstrator; its exact endpoint and clinical value must be agreed before collection.

The protocol would pair repeated experience reports with validated hormone measurements across several cycles, recording sleep, medication, hormonal contraception and relevant context. Reference assays offer a possible initial route while continuous sensors mature. Sampling times, assay performance and missingness must support the chosen horizon. Cohort access, sensor participation and collaborations remain to be agreed.

Separate measurement value from modelling value

Start with persistence and symptom-history forecasts, then assess additional cycle information and hormones. Compare statistical and mechanistic models using the same available inputs. Freeze the evaluation before testing later cycles, retain participant-level separation where the claim concerns new people, and test another cohort before claiming transfer. Report forecast error, calibration, usable coverage and participant burden; thresholds are set before the confirmatory test.

A nested neural study could add structural and diffusion MRI, repeated functional recordings or EEG to estimate how hormonal changes affect individual brain dynamics. Removing hormone inputs and comparing individual with population-average anatomy would test which components earn their acquisition cost. Peripheral concentrations do not directly measure brain exposure or receptor response. The initial hormone-and-experience forecast can establish predictive value; the neural extension tests the mechanistic account.

Fundable package and cost basis

Programme design can deliver a jointly reviewed protocol, minimum measurement specification, access and rights plan, baseline forecasts and a supplier-costed demonstrator budget. Subsequent work needs an explicit Phase I allocation or separate award. The budget must price recruitment and retention, staff, reference assays and sensor calibration, optional imaging, secure processing and independent evaluation. No KAIROS budget or funding agreement is established.

SPRIND's Continuous Hormone Monitoring challenge is a potential source of collaboration, with sensor-development objectives distinct from this several-cycle forecasting study. The design will assess an assay-based route if a shared sensor or data initiative takes longer.

Deliverable and route to use

The first deliverable is an evaluated forecasting demonstrator, documented paired-data protocol and evidence about measurement requirements. Sensor developers, research sponsors and women's-health services are prospective users or purchasers to test. A future application could communicate individual forecasts and uncertainty. Clinical decision support requires separate evaluation of usefulness and safety. Company formation, data rights and commercial deployment remain decisions for a defined product and operating team.

A11. Data access, privacy and the regulatory route

ANPSY will fund data access and operation as programme work. BCOM proposes to define scientific requirements and coordinate delivery; an experienced provider would supply the first secure analysis environment. The choice between central storage, analysis at source or a mixture follows dataset permissions and the required computation. EBRAINS HealthDataCloud is a candidate service whose capacity, cost and fit must be demonstrated, with an alternative provider assessed in parallel.

Make each research task executable

A single access register will identify the custodian, authorised users, relevant participant counts, linked modalities and times, treatment allocation, outcomes, quality and missingness. It will record permission status, lead time, charges and fallback. Separate entries cover model development, independent evaluation and later reuse. A large cohort or an available catalogue does not establish that the required paired records can support the intended test.

Agreements must cover linkage, named development teams, independent code execution, concealed testing, permitted commercial research and publication. They must specify whether features, forecasts, trained weights or executable models may leave the environment. Identifiers and linkage keys normally remain with the source institution. Output review checks permissions and disclosure risk before release.

Assign legal and operating responsibilities

Before processing, institutional legal and privacy teams will document the applicable Article 6 basis and Article 9 condition, controller/processor roles, national requirements, participant information, retention, transfers and any required impact assessment.[32, 33] Research participation, software licensing and permission for secondary data use are separate decisions. Hosting and federation cannot supply missing authorisation.

A funded data lead and steward coordinate custodians, legal teams, the operator and independent evaluator. The operator provides controlled access, project isolation, logging, approved computation, incident response and exit arrangements. Participant-rights requests and corrections must be handled across working copies, releases and retained outputs. Patient and public contributors help assess acceptable use and collection burden.

Preserve scientific usefulness and independent tests

Treat linkable records as personal data unless a documented assessment supports another conclusion. MRI defacing alone is insufficient evidence of anonymity; privacy transformations must also be tested for their effect on the intended analysis.[6] Preserve timing, provenance, units and transformation histories, distinguishing observed, inferred and imputed values.

Development teams cannot access privileged test outcomes. The evaluator controls protected partitions, temporal cutoffs and frozen scoring. An external analyst must reproduce a reference result and an authorised team must execute a model before the service is accepted.

Regulatory work and the delivery gate

For each application, specify intended users, purpose, outputs and influence on care, then obtain a documented assessment of applicable device, AI and clinical-study requirements. Software qualification and classification depend on intended purpose; a research label alone does not settle them.[7]

Programme design costs the access and operating route. Phase I accepts it only with permissions, usable data, tested execution and funded stewardship. Unresolved access triggers a different cohort, narrower claim or deferred component. The conditional vision for wider clinical use is in Section A8.

[6] Schwarz et al., 2021, tests residual recognition and measurement effects; Rubbert et al., 2022, tests pipeline-dependent atrophy estimates.
[7] MDCG 2019-11 rev. 1, June 2025.

A12. Bottlenecks: owners, decisions and alternatives

Each dependency has an accountable role, evidence needed for a decision and a consequence if it remains unresolved. Roles below are responsibilities to appoint during programme design. Months run from programme start; a gate moves when the evidence requires it. The programme maintains a dated register behind this summary.

1. Randomized evidence for treatment selection

Owner: clinical lead, data lead and independent statistician. Define the first decision and required inputs by month 3. Check rights and usable counts by month 6, precision and cost by month 9, and approve an executable existing-data route or funded acquisition by month 12. If neither qualifies, fund only justified forecasting/mechanistic work while withholding treatment-selection qualification. A6 defines the intervention scope and route-selection requirements; the study and costs remain to be agreed.

2. Data permissions and a usable analysis environment

Owner: data lead, custodians, legal teams and operator. Secure permission for development, independent execution and output reuse before committing to a cohort. During Phase I demonstrate the actual workflow, including de-identification, reproducibility and protected evaluation. If rights, utility or service capacity fail, change the dataset or provider, narrow the task or defer it. The operating route is in A11.

3. Mechanistic contribution and informative measurements

Owner: model teams, measurement lead and evaluator. Programme design specifies the component claims and Phase I tests measurement feasibility. Phase II compares equally informed systems and measures the value of neural and experience inputs, personal physiology and explicit coupling. Additional acquisition and computation must earn their burden. Failure to show that contribution limits the mechanistic claim and its next investment; architecture alone confers no clinical-progression preference.

4. Clinically useful treatment differences

Owner: clinical lead and independent statistician. Before Phase II confirmation, estimate achievable rule value and uncertainty, including small-benefit scenarios. If eligible treatments offer little predictable difference, better prognosis cannot create a matching benefit. Revise the population, treatment question or claim on independent evidence; preserve confirmation data from exploratory redesign. The longer-term clinical ambition is in Section A8.

5. Clinical pilot delivery

Owner: clinical operations lead and participating sites. Before recruitment, establish sponsor responsibility, approvals, site capacity, burden and follow-up costs. Timing follows the dependencies shown in A3. Before decision support, test workflow, staff understanding, clinician override and monitoring. Start with qualified sites; add pilot sites within the approved scope using common protocols, external validation and training. If recruitment, safety or usability fails, revise and retest the affected component before expansion. No site participation is presumed.

6. Funding continuity, interests and additional work

Owner: programme/finance lead and independent oversight. Before each award, identify the recipient, priced deliverable, shared costs and continuing operation. Disclose relevant affiliations and interests; recuse conflicted assessors or supplier decision-makers. Record boundaries with VBT/EBRAINS and other grants before contracting overlapping work. Keep KOKORO/KAIROS scope and costs explicit. If the next stage is unfunded, complete the agreed handover and protect data stewardship; further work does not become an automatic liability.

A13. Economic impact: value and assumptions

Better depression care can restore participation in work and daily life as well as relieve suffering. Evaluating these benefits would require later clinical and economic evidence. The OECD models major depressive disorder (MDD) as reducing EU GDP by about 0.7% annually over 2025–2050. Allocating its combined €313 billion annual loss across three conditions by the reported GDP shares gives an approximate MDD baseline of €130 billion per year.[16] This establishes the scale of the problem; the share recoverable through better care needs separate assumptions and evidence.

A transparent long-term scenario

Suppose that improved care reaches half the affected population, delivers the longer-term clinical ambition of 15 additional remissions per 100 patients, and restores 70% of the economic loss associated with each additional remission. The last factor combines persistence of benefit and functional recovery; it is an illustrative assumption. Remission does not necessarily restore functioning fully.[34]

For a constant annual MDD burden B, population reach f, additional remission probability Δr and restoration fraction ρ, the calculation is

$$G = 25Bf\Delta r\rho = 25 \times €130\,\text{bn} \times 0.50 \times 0.15 \times 0.70 \simeq €171\,\text{bn}.$$

This is an illustrative scenario covering 25 years following demonstrated clinical benefit and adoption. It does not predict when those conditions will be met. It requires the improvement to be maintained at population level each year. A 12-week remission endpoint alone does not establish that persistence. Population and real annual burden are held constant; recovered loss is assumed proportional to additional remission.

Adoption, effect size and delivery cost change the result

25-year scenario	Gross output	Delivery cost
50% reach throughout; 15-point gain	€171 bn	€35 bn
Reach rises from 0 to 50%; 15-point gain	€85 bn	€17.5 bn
50% reach throughout; 5-point gain	€57 bn	€35 bn

All rows assume 70% restoration. The gradual scenario increases reach linearly across 25 years. Delivery costs assume an additional €200 per covered person-year and approximately 14 million people with MDD.[16] Reach, effect, restoration and cost are scenario inputs, not achieved outcomes or a proposed product price. Amounts are undiscounted; they exclude research and trial expenditure, and do not model implementation failure. The baseline is OECD-derived; the calculations are ANPSY scenarios.

Evidence needed to assess long-term value

Assessing long-term value would require clinical follow-up measuring remission and relapse, daily functioning, quality of life, employment, absence from work, productivity while working, and incremental care costs. These observations would be needed to replace assumed duration, restoration and delivery costs with estimates. Actual uptake would need to be established across clinics and patient groups.

Gross GDP recovery is distinct from healthcare or government savings. These scenarios do not establish a return on the programme's funding envelope. A later economic evaluation must use compatible price years, discounting and a defined perspective, account for research and deployment costs, and avoid counting the same productivity or care benefit twice.

A14. Why now: a stronger scientific foundation

AI methods, training scale and shared brain resources have expanded. The opportunity is to connect these capabilities in personalised models of brain dynamics, lived experience and treatment response.

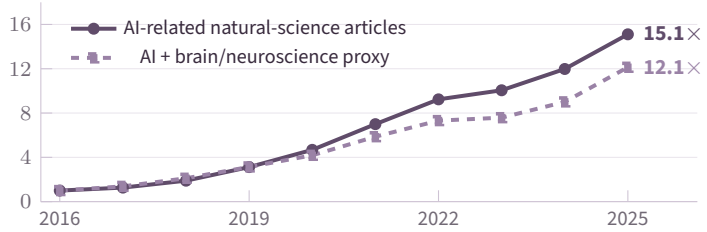
01 AI IN SCIENTIFIC RESEARCH

15.1×

Growth in AI-related natural-science articles: **5,305 to 80,150** per year, 2016–2025.[W1]

The AI-related brain/neuroscience keyword proxy grew **12.1**×: 605 to 7,341 records.[W2]

Annual publications · indexed to 2016 = 1



Different publication corpora and definitions. These indicators measure research activity; their counts are not additive.

02 LARGER AI TRAINING RUNS

≈5× / year

Published growth estimate for training compute in **frontier language models** since 2020.[W3]

90% confidence interval: 4–6× per year. Training compute measures operations used, not intelligence.

Estimated training compute · FLOP, log scale

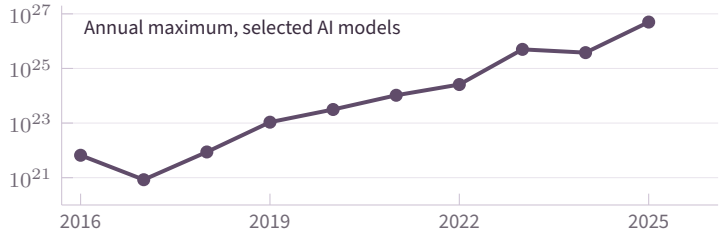


Chart: annual maxima across model domains, including speculative estimates. The 5× headline is a separate language-model trend; no curve is a forecast.

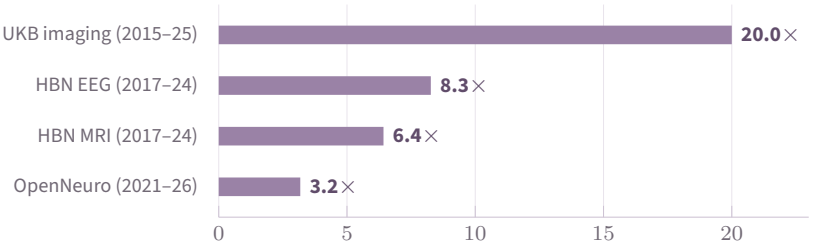
03 SHARED BRAIN RESOURCES

3–20×

Growth between documented release milestones.[W4–6]

UK Biobank: imaging participants. HBN: modality-specific participant datasets. OpenNeuro: public datasets. These are different units and populations.

Scale relative to the earlier release



Endpoints: UKB ≈5,000 → 100,000; HBN EEG 501 → 4,142; HBN MRI 608 → 3,905; OpenNeuro 604 → 1,920. Release dates, not common observation years.

THE OPPORTUNITY FOR ANPSY

Bring these methods, data resources and expertise into one programme: paired measurements, competing personalised models and independent tests. Philanthropy can support shared infrastructure and coordination alongside public funding. Prospective clinical studies will test whether these capabilities improve treatment decisions.

Resource scale does not establish access, paired clinical coverage or clinical benefit. As of 25 Oct 2026, UK Biobank's new applications are paused and UKB-RAP is closed. HBN's 2024 release uses a November 2022 data cutoff.

SOURCES FOR THIS PAGE

Evidence snapshot: 25 October 2026

[W1] Stanford HAI. *AI Index 2026*, Science, Fig. 5.1.1; original chart data.

[W2] Europe PMC: fixed title/abstract queries of PubMed-source records, 2016–2025; snapshot 25 Oct 2026. Query and raw counts archived with the editable source.

[W3] Epoch AI. *Trends in AI* and *AI Models*; frozen data snapshot, 25 Oct 2026 (CC BY 4.0).

[W4] UK Biobank. Past data releases and 2025 major release: released imaging participants, rounded milestones; current access notice.

[W5] Healthy Brain Network. Release 1.0 and release 11.0: cumulative EEG/MRI participant datasets.

[W6] Markiewicz et al. *eLife* 2021;10:e71774; OpenNeuro public-repository snapshot, 25 Oct 2026.

Selected references

[1] M. I. Husain et al. “Measurement-Based Care to Enhance Antidepressant Treatment Outcomes in Major Depressive Disorder: A Randomized Clinical Trial”. In: *JAMA Network Open* 8.9 (2025), e2529427. DOI: 10.1001/jamanetworkopen.2025.29427.

[2] A. Cipriani et al. “A Decision-Support System to Personalize Antidepressant Treatment in Major Depressive Disorder: A Randomized Clinical Trial”. In: *JAMA* 335.14 (2026), pp. 1219–1231. DOI: 10.1001/jama.2026.1327.

[3] K. V. Blake et al. “Genotype-Guided Antidepressant Prescribing for Patients With Depression: A Randomized Clinical Trial”. In: *JAMA Network Open* 9.5 (2026), e2610609. DOI: 10.1001/jamanetworkopen.2026.10609.

[4] Y. Esmaeilian et al. “Predictive neuroimaging biomarkers of major depressive disorder treatment response: An umbrella review”. In: *Psychiatry and Clinical Neurosciences* (2026). DOI: 10.1111/pcn.70053.

[5] I. E. Leaning et al. “From smartphone data to clinically relevant predictions: A systematic review of digital phenotyping methods in depression”. In: *Neuroscience and Biobehavioral Reviews* 158 (2024), p. 105541. DOI: 10.1016/j.neubiorev.2024.105541.

[6] S. H. Siddiqi et al. “The future of brain circuit-targeted therapeutics”. In: *Neuropsychopharmacology* 49 (2024), pp. 179–188. DOI: 10.1038/s41386-023-01670-9.

[7] Human Connectome Project. *HCP Young Adult: 2025 Data Release*. 11, 2025. URL: https://humanconnectome.org/study/hcp-young-adult/data-releases (visited on 10/22/2026).

[8] EBRAINS. *Computing Infrastructure*. URL: https://ebrains.eu/data-tools-services/computing-infrastructure (visited on 10/22/2026).

[9] J. J. Taylor et al. “Connectivity- vs Scalp-Based Targeting of Accelerated Transcranial Magnetic Stimulation for Depression: A Randomized Clinical Trial”. In: *JAMA Psychiatry* (2026), e261100. DOI: 10.1001/jamapsychiatry.2026.1100.

[10] M. A. Jog et al. “Personalized High-Definition Transcranial Direct Current Stimulation for the Treatment of Depression: A Randomized Clinical Trial”. In: *JAMA Network Open* 8.9 (2025), e2531189. DOI: 10.1001/jamanetworkopen.2025.31189.

[11] C. Hesp et al. “Deeply Felt Affect: The Emergence of Valence in Deep Active Inference”. In: *Neural Computation* 33.2 (2021), pp. 398–446. DOI: 10.1162/neco_a_01341.

[12] L. A. Harrison et al. “Resilience phenotypes derived from an active inference account of allostasis”. In: *Frontiers in Behavioral Neuroscience* 19 (2025), p. 1524722. DOI: 10.3389/fnbeh.2025.1524722.

[13] M. Sadeghi et al. “Harnessing multimodal approaches for depression detection using large language models and facial expressions”. In: *npj Mental Health Research* 3 (2024), p. 66. DOI: 10.1038/s44184-024-00112-8.

[14] X. Gu et al. *Harnessing the potential of cellular, animal and computational neuroscience models for mental health*. Landscaping report. Wellcome, 2023. URL: https://wellcome.figshare.com/articles/online_resource/Harnessing_the_potential_of_cellular_animal_and_computational_neuroscience_models_for_mental_health/24321751/1.

[15] C. Zierer, C. Behrendt, and A. C. Lepach-Engelhardt. “Digital biomarkers in depression: A systematic review and call for standardization and harmonization of feature engineering”. In: *Journal of Affective Disorders* 356 (2024), pp. 438–449. DOI: 10.1016/j.jad.2024.03.163.

[16] OECD. *The Economic Case for Preventing Mental Ill Health*. Paris: OECD Publishing, 2026. DOI: 10.1787/16668f16-en.

[17] Eurostat. *EU population continued to grow in 2025*. 10, 2026. URL: https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20260710-3 (visited on 10/21/2026).

[18] G. Ruffini et al. “The Algorithmic Agent Perspective and Computational Neuropsychiatry: From Etiology to Advanced Therapy in Major Depressive Disorder”. In: *Entropy* 26.11 (2024), p. 953. DOI: 10.3390/e26110953.

[19] P. Sanz Leon et al. “The Virtual Brain: a simulator of primate brain network dynamics”. In: *Frontiers in Neuroinformatics* 7 (2013), p. 10. DOI: 10.3389/fninf.2013.00010.

[20] R. Sanchez-Todo et al. “A physical neural mass model framework for the analysis of oscillatory generators from laminar electrophysiological recordings”. In: *NeuroImage* 270 (2023), p. 119938. DOI: 10.1016/j.neuroimage.2023.119938.

[21] G. Deco et al. “Whole-Brain Multimodal Neuroimaging Model Using Serotonin Receptor Maps Explains Non-linear Functional Effects of LSD”. In: *Current Biology* 28.19 (2018), 3065–3074.e6. DOI: 10.1016/j.cub.2018.07.083.

[22] N. Reimers and I. Gurevych. “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks”. In: *Proceedings of EMNLP-IJCNLP*. 2019, pp. 3982–3992. DOI: 10.18653/v1/D19-1410.

[23] O. N. E. Kjell et al. “Semantic measures: Using natural language processing to measure, differentiate, and describe psychological constructs”. In: *Psychological Methods* 24.1 (2019), pp. 92–115. DOI: 10.1037/met0000191.

[24] H. S. Mayberg et al. “Deep Brain Stimulation for Treatment-Resistant Depression”. In: *Neuron* 45.5

(2005), pp. 651–660. DOI: 10.1016/j.neuron.2005.02.014.

[25] M. D. Fox et al. "Efficacy of Transcranial Magnetic Stimulation Targets for Depression Is Related to Intrinsic Functional Connectivity with the Subgenual Cingulate". In: *Biological Psychiatry* 72.7 (2012), pp. 595–603. DOI: 10.1016/j.biopsych.2012.04.028.

[26] K. Kalafatakis et al. "Ultradian rhythmicity of plasma cortisol is necessary for normal emotional and cognitive responses in man". In: *Proceedings of the National Academy of Sciences* 115.17 (2018), E4091–E4100. DOI: 10.1073/pnas.1714239115.

[27] D. S. Avila-Varela et al. "Whole-brain dynamics across the menstrual cycle: the role of hormonal fluctuations and age in healthy women". In: *npj Women's Health* 2 (2024), p. 8. DOI: 10.1038/s44294-024-00012-4.

[28] M. D. Wilkinson et al. "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3, 160018 (2016). DOI: 10.1038/sdata.2016.18.

[29] T. Gneiting and A. E. Raftery. "Strictly Proper Scoring Rules, Prediction, and Estimation". In: *Journal of the American Statistical Association* 102.477 (2007), pp. 359–378. DOI: 10.1198/016214506000001437.

[30] M. Dudík, J. Langford, and L. Li. *Doubly Robust Policy Evaluation and Learning*. 2011. URL: https://arxiv.org/abs/1103.4601 (visited on 09/16/2026).

[31] B. Vasey, M. Nagendran, B. Campbell, et al. "Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI". In: *BMJ* 377 (2022), e070904. DOI: 10.1136/bmj-2022-070904.

[32] European Parliament and Council of the European Union. *Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation)*. OJ L 119, 4.5.2016, p. 1. 27, 2016. URL: https://eur-lex.europa.eu/eli/reg/2016/679/oj (visited on 10/08/2026).

[33] European Data Protection Board. *Guidelines 07/2020 on the concepts of controller and processor in the GDPR*. Version 2.1. Final version adopted 7 July 2021; minor corrections 20 September 2022. 7, 2021. URL: https://www.edpb.europa.eu/our-work-tools/our-documents/guidelines/guidelines-072020-concepts-controller-and-processor-gdpr_en (visited on 10/08/2026).

[34] R. Schwarz et al. "Functioning in patients with major depressive disorder in remission: A systematic review and meta-analysis". In: *Journal of Affective Disorders* 363 (2024), pp. 112–123. DOI: 10.1016/j.jad.2024.07.054.